Measures of Disease Occurrence - Part 2

APHS 20203 • Epidemiology & Biostatistics

Course Learning Objectives

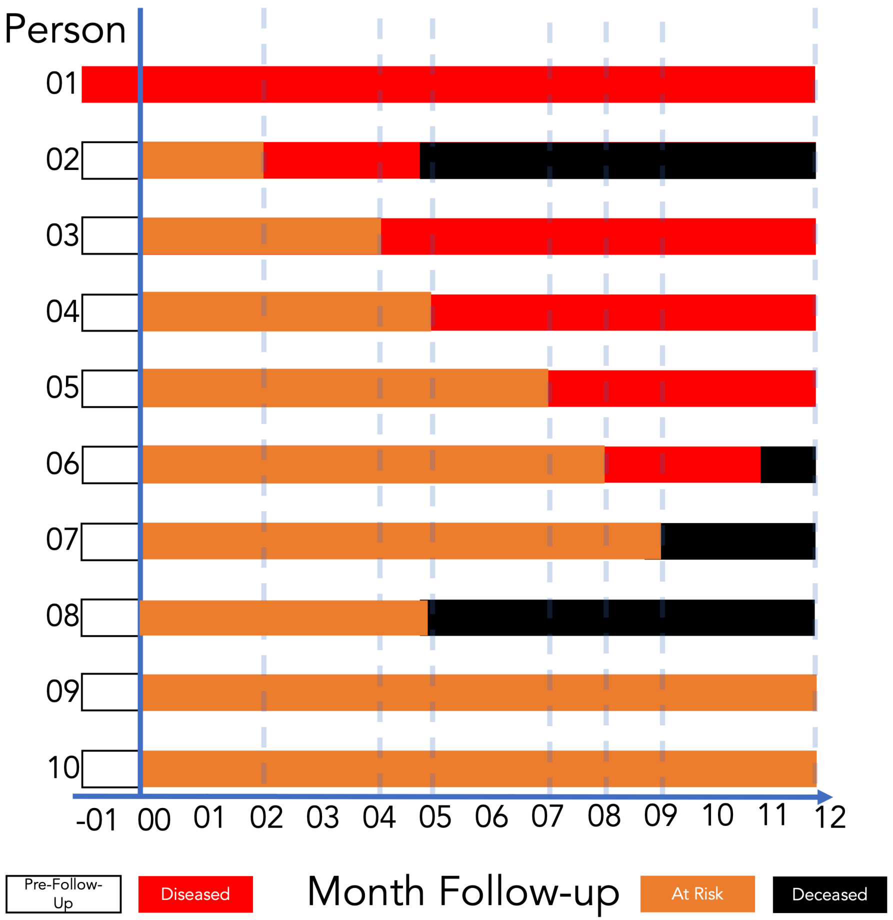
1. Define epidemiology and key associated terminology, especially confounding.
2. Distinguish among association, prediction, and causation.
3. Explain the Rothman model of sufficient-component causes (RMSCC) and the concept of multifactorial causation.
4. Explain the limitations of anecdotal evidence in drawing epidemiologic conclusions.
5. Explain why disseminating risk information is sometimes necessary but rarely sufficient to improve population health.

Lecture Learning Objectives

By the end of this lesson, you'll be able to:

1. Define counts and rates.
2. Distinguish crude, specific, adjusted rates.
3. Compute and interpret prevalence and incidence.

Review: Choose the Snapshot



With your group:

1. Choose **month 2, 6, or 10**.
2. Count the people living with disease.
3. Count the living members of the population.
4. Calculate and interpret the prevalence proportion.

Be ready to explain your numerator and denominator.

Prevalence Describes a State

Disease present

Who has disease?

A **state** observed at a moment or over a period

Incidence Describes a Transition

Disease absent → **Disease present**

At risk

during follow-up

Who developed disease?

Incidence counts **new transitions** during a specified period.

One-Time vs. Recurrent Events

We can measure the incidence of:

- **One-time events**, such as Parkinson's disease or a first cancer diagnosis
- **Recurrent events**, such as respiratory infections during a calendar year

Incidence Proportion and Risk

- Incidence proportion and “risk” are sometimes used interchangeably in epidemiology.
- Risk can be ambiguous. Incidence proportion is less so.

Incidence Measures

Incidence Count

- A count of new occurrences of a condition in a population at risk for the occurrence during a given time frame.
- Range: 0 to infinity.

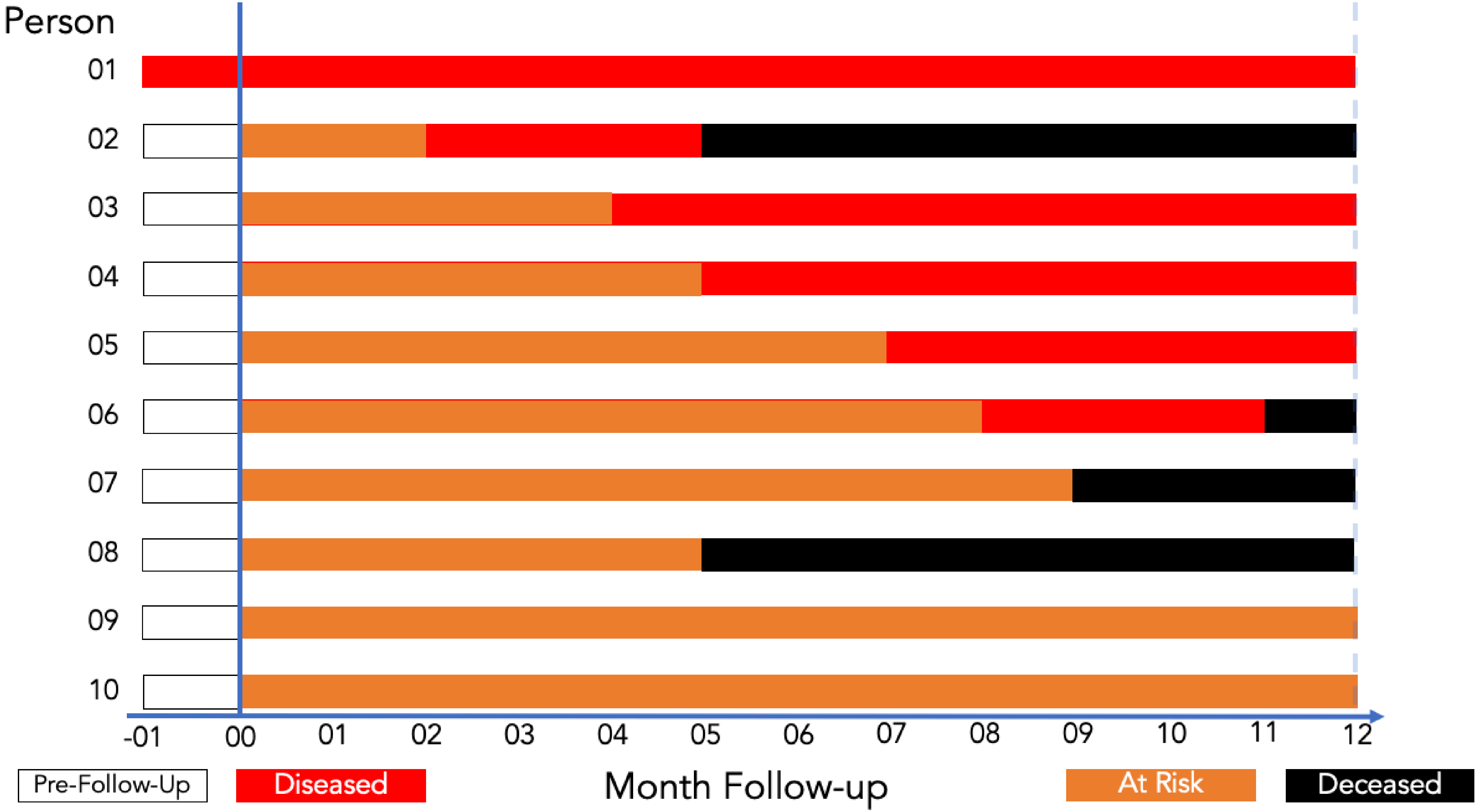
Incidence Proportion

- The proportion of the population that experiences a new occurrence of the condition among those at risk of experiencing a new occurrence during a given time frame.
- Like all proportions, it falls between 0 and 1.

Incidence Proportion: Formula

$$\text{Incidence proportion} = \frac{\text{Count of new occurrences}}{\text{Population at risk}}$$

Follow-Up Experience and New Occurrences



Source: Lash TL, VanderWeel TJ, Haneuse S, Rothman KJ. *Modern Epidemiology*. fourth. Wolters Kluwer; 2021.

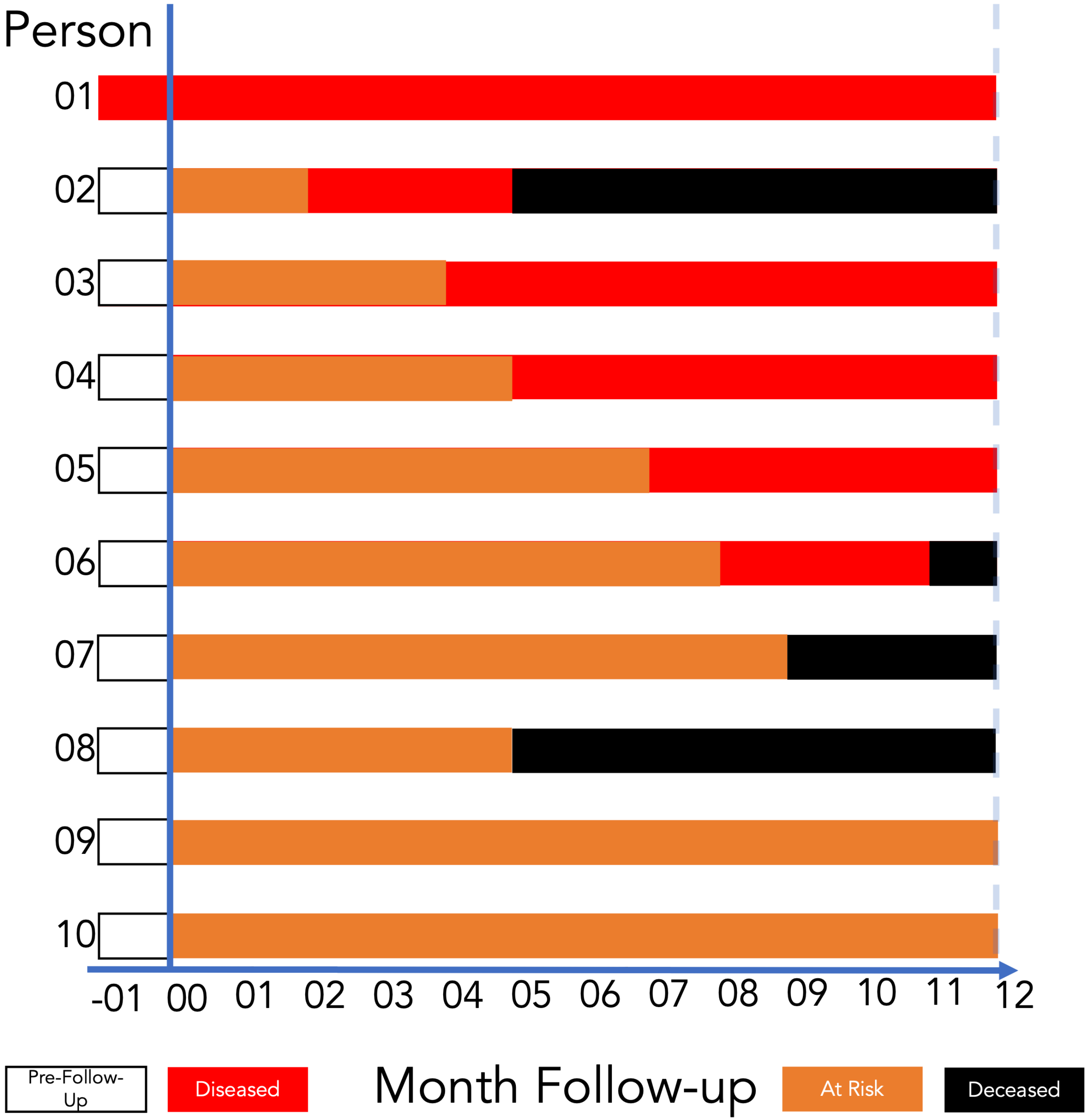
Incidence Proportion Over 12 Months

$$\frac{\text{Count of new occurrences}}{\text{Population at risk}}$$

$$\frac{5}{9}$$

$$\approx 0.56 \text{ or } 56\%$$

Interpreting Incidence Proportion



Among the members of the population, the incidence of disease over 12 months of follow-up was **0.56**.

Fifty-six percent of the population developed disease over 12 months of follow-up.

Incidence Odds: Formula

$$\text{Incidence odds} = \frac{\text{Incidence proportion}}{1 - \text{Incidence proportion}}$$

Incidence Odds Over 12 Months

$$\frac{\text{Incidence proportion}}{1 - \text{Incidence proportion}}$$

$$\frac{0.56}{1 - 0.56}$$

$$\approx 1.27$$

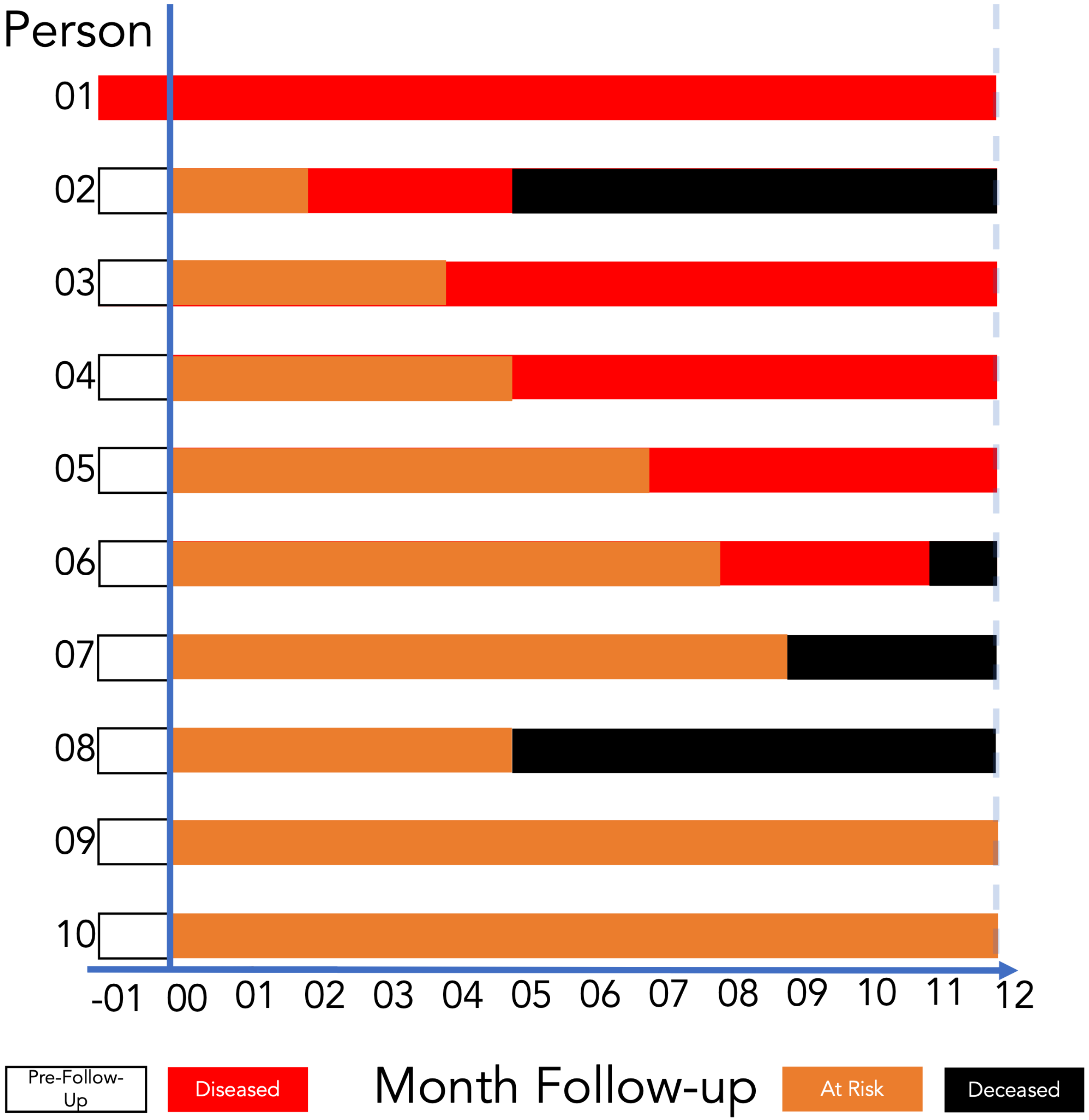
Exact Incidence Odds Over 12 Months

$$\frac{\text{Incident cases}}{\text{Non-incident cases}}$$

$$5{:}4 \text{ or } \frac{5}{4}$$

$$= 1.25$$

Interpreting Incidence Odds



Among the members of the population, the odds of incident disease over 12 months of follow-up were **1.25**.

For every 5 people who developed incident disease over 12 months of follow-up, 4 people did not.

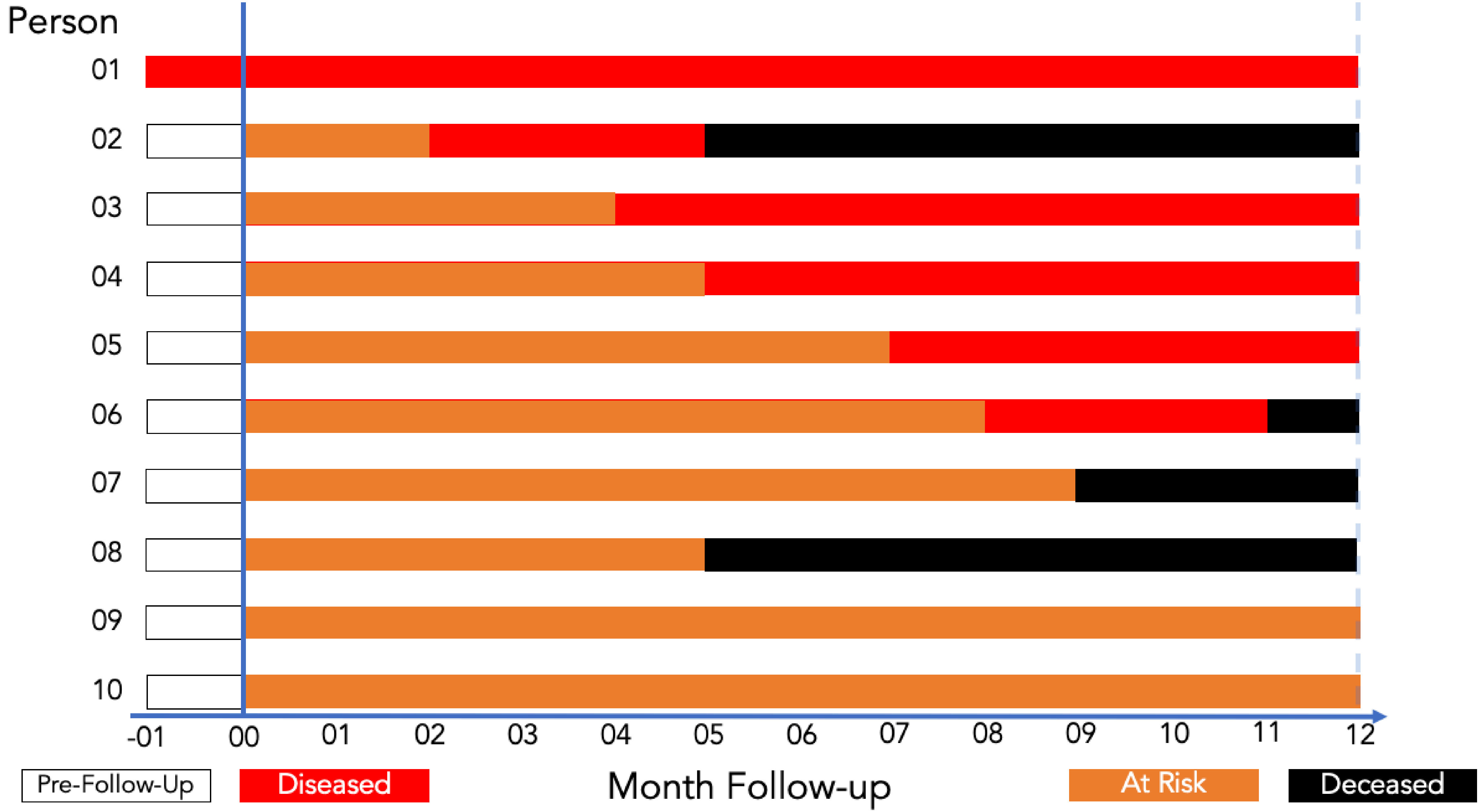
Incidence Rate

- A measure of the average intensity at which an event occurs in people's experience over time.
- For a population followed from baseline, it is the number of cases divided by the person-time at risk accumulated by the population.
- Range: 0 to infinity.
- **Not a proportion.**
- The denominator is person-time at risk
- The resulting rate has units of inverse/reciprocal time (time^{-1}).
- The numeric value depends on the selected time unit and therefore has no interpretation by itself.

Person-Time

- **Person-time** is the amount of time observed for all people under study while they are at risk of experiencing an incident outcome.
- If a woman is followed for 1 month, then she experiences (and contributes to analysis) 1 person-month of time.
- If 10 women are followed for 1 year, they contribute $10 \times 1 = 10$ person-years.
- If 1 woman is followed for 10 years, she also contributes $1 \times 10 = 10$ person-years.

Follow-Up Experience and Time at Risk



Source: Lash TL, VanderWeel TJ, Haneuse S, Rothman KJ. *Modern Epidemiology*. fourth. Wolters Kluwer; 2021.

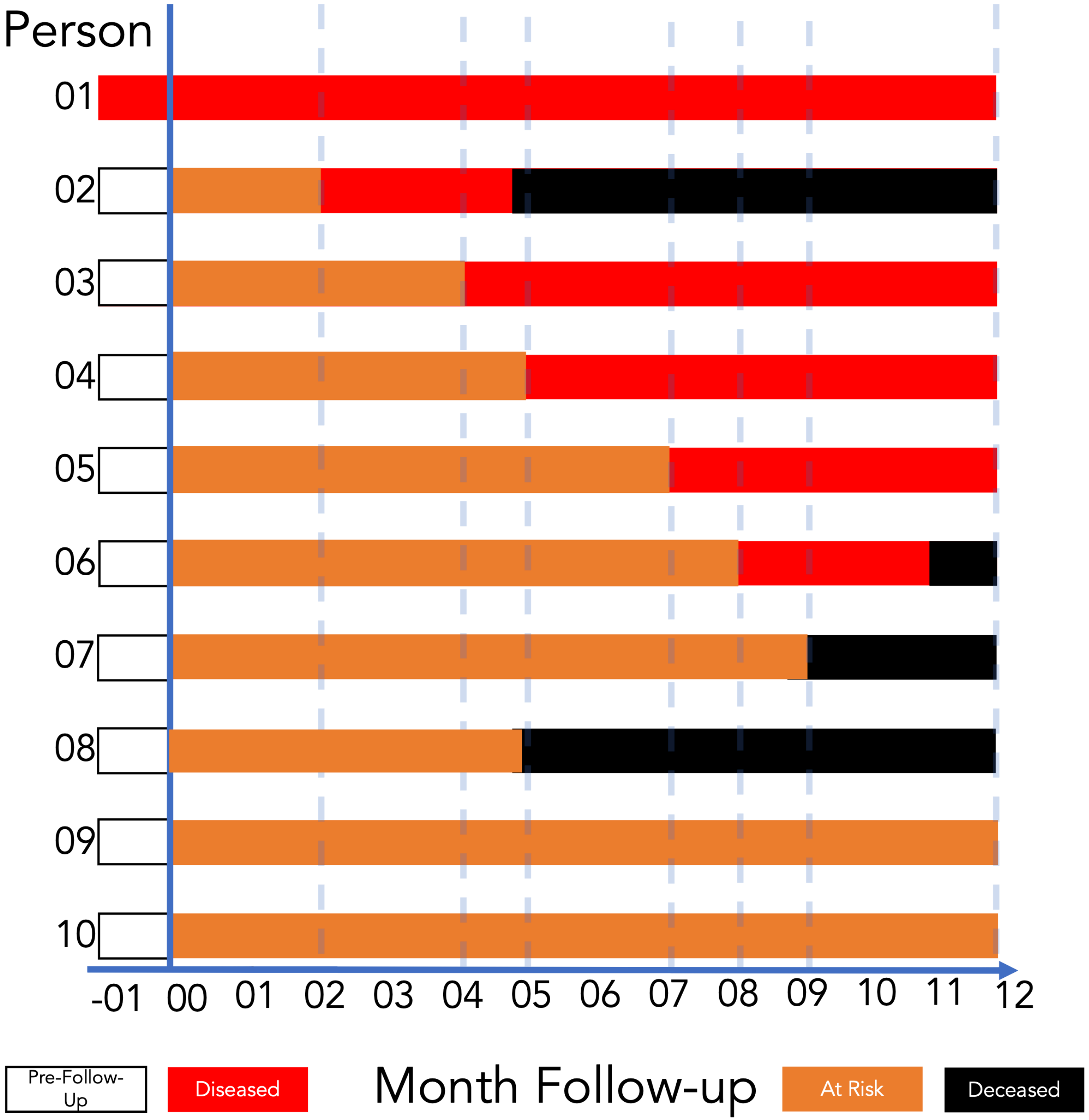
Calculating Person-Time

$$\sum_{\text{people}} \text{Time at risk}$$

$$= 0 + 2 + 4 + 5 + 7$$
$$+ 8 + 9 + 5 + 12 + 12$$

$$= 64 \text{ person-months}$$

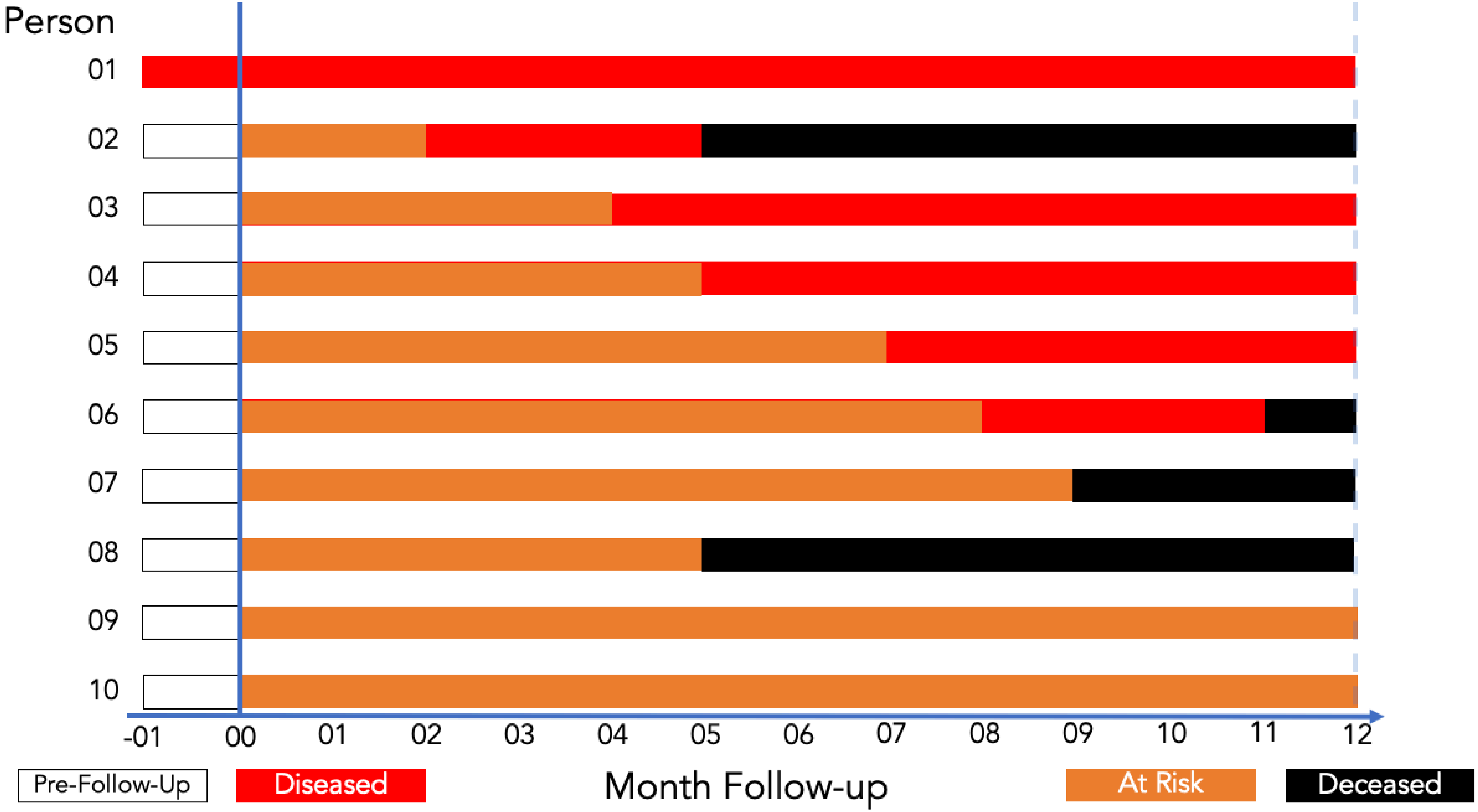
Interpreting Person-Time



The population accumulated **64 person-months at risk** during 12 months of follow-up.

The same amount is **5.33 person-years at risk** during 12 months of follow-up.

Follow-Up Experience for the Incidence Rate



Source: Lash TL, VanderWeel TJ, Haneuse S, Rothman KJ. *Modern Epidemiology*. fourth. Wolters Kluwer; 2021.

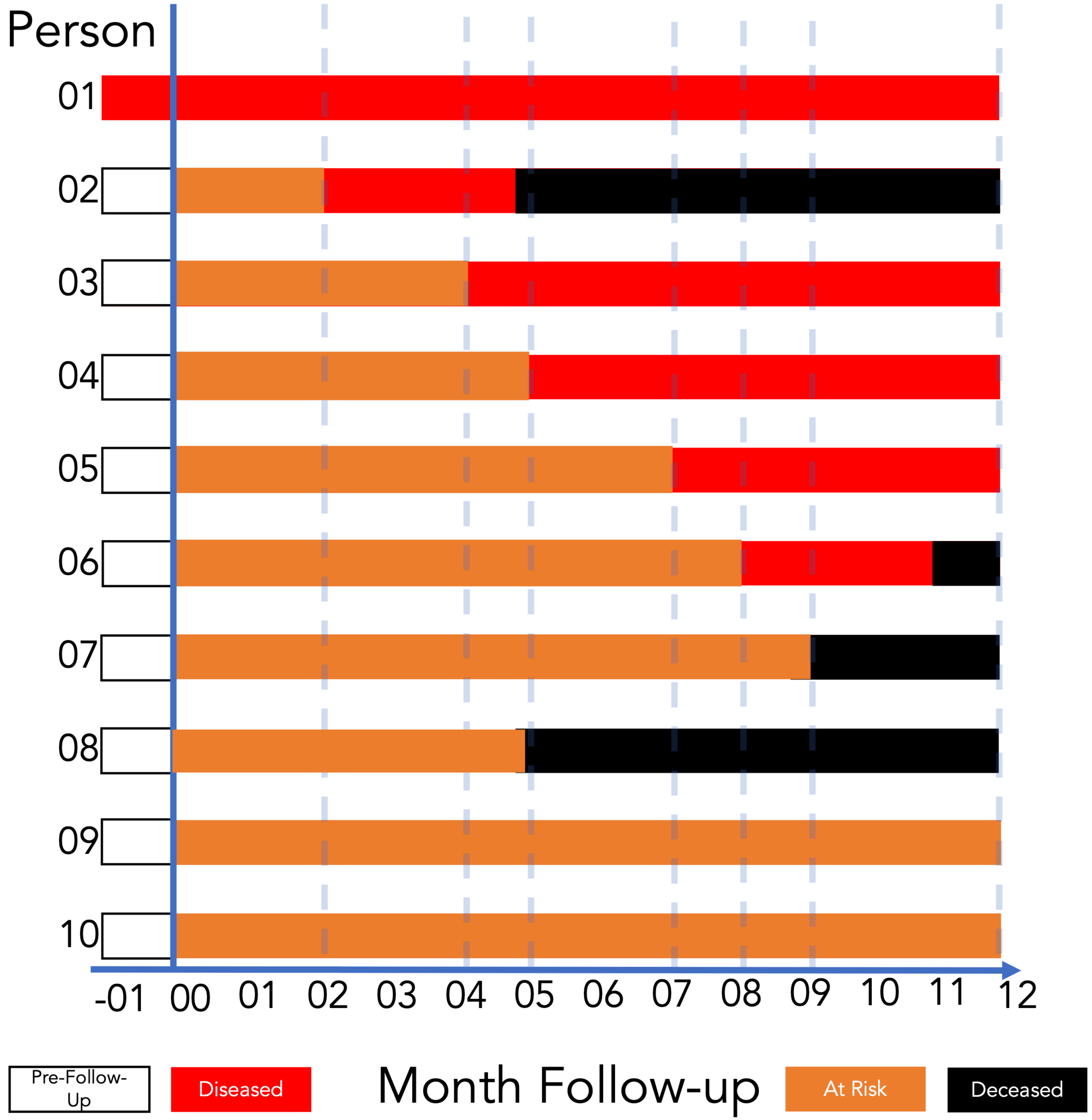
Incidence Rate: Calculation

$$\frac{\text{Count of new occurrences}}{\text{Person-time at risk}}$$

$$= \frac{5}{\text{64 person-months}}$$

$$\approx \text{7.8 per 100 person-months}$$

Interpreting the Incidence Rate



The incidence rate of disease among the members of the population was **5 cases per 64 person-months** during 12 months of follow-up.

Equivalently, the incidence rate was **7.8 cases per 100 person-months** during 12 months of follow-up.

Inverse/Reciprocal Time (time^{-1})

- If a study follows 100 people for 5 years, what is the total person-time?
  - The total person-time is 500 person-years.
- If 10 people experience the event of interest, what is the incidence rate?
  - The incidence rate is: $\frac{10 \text{ cases}}{500 \text{ person-years}} = 0.02 \text{ cases per person-year}$
  - The unit is 0.02 time^{-1}. This reciprocal time unit means that, on average, 2% of the population at risk will experience the event each year.
- But why is it called "reciprocal time" and why is it written as time^{-1}?

Why Is It Called Inverse/Reciprocal Time?

- It is called reciprocal time and written as time^{-1} because of basic algebra and the rules of mathematical exponents.
- In mathematics, the reciprocal of any number or variable is just 1 divided by that variable.
  - The reciprocal of 5 is $\frac{1}{5}$
  - The reciprocal of Time is $\frac{1}{\text{Time}}$.
  - When we calculate an incidence rate, the formula is: $\frac{\text{Count of new occurrences}}{\text{Person-time at risk}}$
- Because the unit of time sits in the **denominator** (the bottom of the fraction), we are multiplying the number of cases by the reciprocal of time.

Why the Negative Exponent (time^{-1})?

- In algebra, a negative exponent is a shorthand way of writing a fraction where the variable is on the bottom. The standard rule is: $\frac{1}{x} = x^{-1}$
- If our unit of time is years, then: $\frac{1}{\mathrm{year}} = \mathrm{year}^{-1}$
- So, when we see a rate written as $0.02\ \mathrm{time}^{-1}$, it literally means 0.02 per year or $\frac{0.02}{\mathrm{year}}$.
- Using the negative exponent is simply the standard scientific way to write “per year” on a single line without needing to draw a fraction bar.

Mortality Rates: Incidence of Death

Death is another **new event** we can count over time.

$$\text{Crude mortality rate} = \frac{\text{Deaths during the year}}{\text{Midyear population}} \times 1{,}000$$

Hypothetical town: 80 deaths during one year; 10,000 residents at midyear.

$$\frac{80}{10{,}000} \times 1{,}000 = 8 \text{ deaths per } 1{,}000 \text{ residents that year}$$

- **Crude:** all causes of death, across the whole population.
- **Specific:** narrow the cause or population, such as cancer deaths or deaths among adults ages 65+.

Natality Rates: Counting Births

Natality refers to births. We again relate new events to a population over time.

$$\text{Crude birth rate} = \frac{\text{Live births during the year}}{\text{Midyear population}} \times 1{,}000$$

Hypothetical town: 120 live births; 10,000 residents at midyear.

$$\frac{120}{10{,}000} \times 1{,}000 = 12 \text{ live births per } 1{,}000 \text{ residents that year}$$

- This describes births in relation to the **whole population**.
- The **general fertility rate** uses women ages 15–44 as its denominator, by U.S. statistical convention.

A Familiar Name Can Hide a Different Denominator

Always check **what is being divided by what**.

Measure	Numerator	Denominator
Case-fatality proportion	Deaths from the disease among a defined group of cases	All cases in that group
Infant mortality rate	Deaths before age 1 during a year	Live births during that year

Hypothetical examples:

- **Case fatality:** 3 deaths among 100 cases followed through illness → **3%** of those cases died from the disease.
- **Infant mortality:** 6 infant deaths and 1,200 live births → **5 deaths before age 1 per 1,000 live births** that year.

Why Adjust a Rate?

A town with more older adults can have a higher crude death rate **even when its age-specific rates are the same**.

Hypothetical example: one year, 10,000 residents in each town.

Age group	Town A population	Town B population	Deaths per 1,000 in both towns
Under 65	9,000	5,000	2
65+	1,000	5,000	20

- **Town A:** 18 + 20 = 38 deaths → **3.8 per 1,000**.
- **Town B:** 10 + 100 = 110 deaths → **11 per 1,000**.

What would the comparison look like if both towns had the **same age mix**?

Age Adjustment: Use the Same Age Mix

AI-created draft — needs review

Direct age adjustment takes a weighted average of age-specific rates using the **same age weights** for every population.

For our example, choose a standard population that is **80% under 65** and **20% ages 65+**.

$$\text{Adjusted rate} = (0.80 \times 2) + (0.20 \times 20) = 5.6$$

Both towns have an adjusted rate of **5.6 deaths per 1,000 standard population per year**.

- This is the rate expected if each town had the standard age mix.
- Age adjustment supports comparisons; it does **not** remove every difference or bias between populations.

Incidence Rates

Why do we sometimes calculate incidence rates instead of incidence proportions, which are arguably more intuitive for most people?

Respond here

Review: Ratios, Proportions, and Percentages

In a survey conducted today, 12 of 40 students report riding a bicycle to campus; 28 do not. Calculate the ratio of bicycle riders to nonriders, the proportion who ride, and the percentage who ride.

- Riders to nonriders = 12/28 = 3/7, or 3:7: three riders for every seven nonriders.
- The proportion who ride is 12/(12+28) = 12/40 = 0.30.
- The percentage is 0.30 × 100 = 30%.
- The ratio compares riders with nonriders; the proportion and percentage compare riders with all surveyed students.
- All three descriptions refer to this survey today.

Review: Prevalence and Incidence

On January 1, 2026, 20 of 200 adults had hypertension. During 2026, 36 of the remaining 180 develop it. Everyone is followed for the full year. What are the January 1 prevalence proportion and the 2026 incidence proportion, respectively?

A. 10% and 18%

B. 10% and 20%

C. 28% and 20%

D. 20% and 10%

Respond here

Review: Person-Time and Rates

In a 6-month study, four athletes contribute 2, 4, 6, and 6 months at risk of a first ankle injury. The first two are injured at months 2 and 4; the others remain injury-free. Calculate person-time and the incidence rate per 100 person-months.

Respond here

Review: Interpret the Odds

In a survey conducted today, 30 residents have asthma and 90 do not. Which statement correctly interprets the prevalence odds of asthma?

A. There is 1 resident with asthma for every 3 without asthma.

B. There is 1 resident with asthma for every 4 without asthma.

C. There are 3 residents with asthma for every 1 without asthma.

D. There is 1 resident with asthma for every 3 residents in total.

Respond here

Review: Convert Odds to Probability

Among people tested at a clinic today, the odds of a positive test are 3:2. What probability of a positive test corresponds to these odds? Show the calculation and express the answer as a percentage.

Respond here

Review: Compare Two Odds

Two groups are surveyed for insomnia today. The prevalence odds are 2.0 in Group A and 0.5 in Group B. Which statement correctly compares the groups?

A. Group A has twice Group B's odds of insomnia.

B. Group A has four times Group B's probability of insomnia.

C. Group A has four times Group B's odds of insomnia.

D. Group A has one-fourth of Group B's odds of insomnia.

Respond here

Review: Why Adjust Rates?

Two counties have the same death rates within each age group in 2026, but County A has a larger share of older residents and a higher crude death rate. What does applying the same standard age distribution to both counties accomplish?

A. It changes the actual number of deaths recorded in each county.

B. It removes every possible source of bias from the comparison.

C. It makes each county's age-specific death rates equal to its crude rate.

D. It compares summary death rates using the same age mix.

Respond here

Review: Mortality Rates

During 2026, a county records 360 deaths from all causes and has a midyear population of 90,000. Which statement correctly reports its crude mortality rate per 1,000 residents?

A. 0.4 deaths per 1,000 residents during 2026.

B. 4 deaths per 1,000 residents during 2026.

C. 4 deaths per 1,000 people who died during 2026.

D. 40 deaths per 1,000 residents during 2026.

Respond here

Review: Birth and Fertility Rates

During 2026, a town records 210 live births. Its midyear population is 14,000, including 3,000 women ages 15–44. Calculate the crude birth rate and the U.S. general fertility rate, each per 1,000. Explain why the answers differ.

Respond here

References

Westreich, Daniel. 2019. *Epidemiology by Design: A Causal Approach to the Health Sciences*. Oxford University Press.